

LocationCircle — Closed-Group Location Sharing Platform

Document Revision History

Version	Date	Author	Notes
1.0	17 May 2026	Intern / Project Team	Initial draft

1. Executive Summary

LocationCircle is a closed-group location-sharing web application (progressively wrappable as a mobile app) that enables trusted circles of friends and family to see each other's real-time geographic positions in a private, invite-only environment. The platform addresses the growing need for safe, intimate location sharing without the privacy concerns of public social networks.

The application will be accessible via modern web browsers and, through WhatsApp integration, usable by members who prefer not to install additional apps. Accessibility for low-vision users is a first-class requirement, ensured through WCAG 2.1 AA compliance and built-in text-to-speech.

2. Project Scope

2.1 In Scope

- Closed-group creation, invitation, and membership management
- Real-time location sharing (GPS / network-based geolocation)
- Group Head designation, including Temporary Head assignment
- Main dashboard — member list with locality, distance, ETA (car & transit)
- Drill-down map view with member icons / profile pictures
- In-app call-to-phone action from the member list
- Google OAuth 2.0 authentication
- WhatsApp Business API integration (location updates, bot interface)
- Accessibility: WCAG 2.1 AA, screen-reader support, text-to-speech readout
- Account settings — phone number, WhatsApp linking, group management
- Responsive PWA shell ready for Capacitor/React Native wrapping

2.2 Out of Scope (v1)

- Native iOS / Android store submissions
- Route history playback or location analytics
- Payments or subscription billing
- Integration with platforms other than WhatsApp and Google

3. Stakeholders

Role	Description	Interest
Group Head	Administrator of a LocationCircle group	Manage members, set Temporary Head, monitor group
Temporary Head	Delegated head during Head's unavailability	Assume head visibility & limited admin rights
Group Member	Invited participant sharing their location	See group positions; contact members; use WhatsApp interface

Role	Description	Interest
Internship Team	Developers & designers building the app	Deliver on requirements within internship timeline
Project Supervisor	Mentor / evaluator of the internship project	Quality, learning outcomes, timely delivery

4. Functional Requirements

4.1 Authentication & Onboarding

ID	Priority	Requirement	MoSCoW
FR-AUTH-01	High	Users must sign in exclusively via Google OAuth 2.0; no username/password creation.	Must
FR-AUTH-02	High	On first login the user is prompted to set a display name, upload a profile picture, and enter a mobile phone number.	Must
FR-AUTH-03	Medium	Users may link their WhatsApp account (phone number verification via OTP) in Account Settings.	Should
FR-AUTH-04	Low	Session tokens refresh silently; users are not re-prompted unless idle for 30 days.	Could

4.2 Group Management

ID	Priority	Requirement	MoSCoW
FR-GRP-01	High	Any authenticated user can create a new LocationCircle group; the creator automatically becomes the Group Head.	Must
FR-GRP-02	High	The Group Head can invite new members via shareable link or direct Google-account lookup.	Must
FR-GRP-03	High	The Group Head can designate any member as the new permanent Group Head; the previous Head reverts to member.	Must
FR-GRP-04	High	The Group Head can assign a Temporary Head with a specified expiry date/time; on expiry the original Head is automatically reinstated.	Must
FR-GRP-05	Medium	The Temporary Head has the same real-time visibility prominence as the Group Head but cannot change permanent head assignments.	Should
FR-GRP-06	Medium	Members can leave a group; the Group Head can remove a member.	Should
FR-GRP-07	Low	A user may belong to multiple groups simultaneously.	Could

4.3 Location Sharing

ID	Priority	Requirement	MoSCoW
FR-LOC-01	High	The app requests device geolocation permission; location is updated continuously while the app is in the foreground.	Must
FR-LOC-02	High	The Group Head's (and active Temporary Head's) location is displayed prominently / refreshed immediately on all members' dashboards.	Must

ID	Priority	Requirement	MoSCoW
FR-LOC-03	High	Each member's entry on the dashboard shows: closest town / locality name, approximate straight-line distance from the viewing user, and ETA by car and by public transport.	Must
FR-LOC-04	Medium	ETA data is fetched from the Google Maps Distance Matrix API (or equivalent).	Should
FR-LOC-05	Medium	Members can pause their own location sharing at any time; their entry shows 'Location paused'.	Should
FR-LOC-06	Low	Location updates are pushed via WebSocket / Server-Sent Events for near-real-time accuracy.	Should

4.4 Main Dashboard (Member List)

ID	Priority	Requirement	MoSCoW
FR-DASH-01	High	The landing page after login is a scrollable, accessible list of all group members sorted by proximity to the viewer.	Must
FR-DASH-02	High	The Group Head entry is visually distinct (pinned to top, badge icon, highlighted row).	Must
FR-DASH-03	High	Each list item includes: profile picture / avatar, display name, locality, distance, ETA car, ETA transit, and a 'Call' action button.	Must
FR-DASH-04	High	The 'Call' button initiates a mobile phone call to the member's registered number via the device's native dialler (tel: URI).	Must
FR-DASH-05	High	The entire list is fully keyboard-navigable and screen-reader compatible (ARIA roles, labels, live regions for location updates).	Must
FR-DASH-06	High	A 'Read Aloud' control reads each list item using the Web Speech API (TTS), useful for low-vision or hands-free use.	Must
FR-DASH-07	Medium	Tapping / clicking a list item opens the Drill-Down Map view for that member.	Must

4.5 Drill-Down Map View

ID	Priority	Requirement	MoSCoW
FR-MAP-01	High	Selecting a member opens a full-screen map (Google Maps JS SDK or Mapbox) centred on that member.	Must
FR-MAP-02	High	All group members are shown as markers using their profile picture inside a circular frame; the Head marker is visually distinct.	Must
FR-MAP-03	Medium	The viewing user's own position is shown as a separate 'You are here' marker.	Must
FR-MAP-04	Low	Map supports standard pinch-zoom, pan, and satellite/street toggle.	Should

ID	Priority	Requirement	MoSCoW
FR-MAP-05	Low	An accessible text summary of marker positions is available via a hidden ARIA live region.	Could

4.6 Account Settings

ID	Priority	Requirement	MoSCoW
FR-SET-01	High	Users can update display name, profile picture, and phone number.	Must
FR-SET-02	High	Group Head can invite members, remove members, transfer permanent headship, and assign/revoke Temporary Head from this screen.	Must
FR-SET-03	High	Members can link / unlink their WhatsApp account by entering the phone number and verifying via OTP.	Must
FR-SET-04	Medium	Users can configure notification preferences (push, WhatsApp, email).	Should
FR-SET-05	Low	Users can delete their account and all associated location data (GDPR right to erasure).	Must

4.7 WhatsApp Integration

ID	Priority	Requirement	MoSCoW
FR-WA-01	High	Members with a linked WhatsApp account receive location-update notifications via WhatsApp Business API messages.	Must
FR-WA-02	High	A WhatsApp bot interface allows members to query 'Where is [Name]?' and receive a text + Google Maps link reply.	Should
FR-WA-03	Medium	The bot supports commands: LIST (all members), HEAD (head's location), PAUSE (pause sharing), RESUME.	Should
FR-WA-04	Low	Bot responses are formatted for screen readers and plain-text environments.	Could

5. Non-Functional Requirements

5.1 Accessibility

- WCAG 2.1 Level AA compliance across all pages.
- Minimum contrast ratio 4.5:1 for normal text; 3:1 for large text.
- All interactive elements reachable and operable via keyboard.
- Screen-reader tested on NVDA (desktop) and TalkBack / VoiceOver (mobile).
- Text-to-speech (Web Speech API) available on every list view.
- Font size adjustable without loss of content or functionality (up to 200%).

5.2 Performance

- Initial page load < 3 s on a 4G connection.
- Location list refreshes within 5 s of a member moving.
- Map tile load and marker render < 2 s after drill-down tap.
- API response time (distance/ETA) < 1.5 s at p95.

5.3 Security & Privacy

- All data in transit encrypted via TLS 1.2+.
- Location data stored encrypted at rest; purged within 24 h of member removal.
- Google OAuth tokens stored in secure HttpOnly cookies; no localStorage.
- WhatsApp OTP verification expires after 10 minutes.
- Compliance with GDPR / applicable local data-protection regulations.

5.4 Reliability & Availability

- Target uptime: 99.5% monthly (excluding scheduled maintenance).
- Graceful degradation: if location services fail, last-known position displayed with timestamp.
- Offline-first PWA caching of the member list for read-only access without connectivity.

5.5 Scalability

- Architecture must support up to 50 concurrent groups of up to 30 members each in v1.
- Backend stateless services to allow horizontal scaling.

5.6 Internationalisation

- UI text externalised into i18n resource files to support future language additions.
- Distances displayed in km or miles based on user locale.

6. System Architecture Overview

The following describes the recommended high-level architecture for the internship build. Final technology choices are subject to team skills and supervisor approval.

Layer	Component	Recommended Technology
Frontend	Web App (PWA)	React + TypeScript, Tailwind CSS, Vite
Frontend	Maps	Google Maps JavaScript SDK / Mapbox GL JS
Frontend	TTS / A11y	Web Speech API, react-aria / Headless UI
Backend	API Server	Node.js + Express / FastAPI (Python)
Backend	Real-time updates	Socket.io / Server-Sent Events
Backend	WhatsApp Bot	WhatsApp Business Cloud API (Meta)
Auth	Identity	Google OAuth 2.0 (via Passport.js or Firebase Auth)
Data	Primary DB	PostgreSQL (members, groups, settings)
Data	Location store	Redis (ephemeral real-time positions)
Infra	Hosting	Google Cloud Run / Render / Railway (internship budget-friendly)
Infra	Mobile wrap	Capacitor (post-MVP)

7. User Roles & Permissions Matrix

Permission	Member	Temp Head	Group Head
View member list & map	✓	✓	✓
Share own location	✓	✓	✓
Pause own location	✓	✓	✓
Call a member	✓	✓	✓
Use WhatsApp bot	✓	✓	✓
Invite new members	✗	✗	✓
Remove members	✗	✗	✓
Transfer permanent headship	✗	✗	✓
Assign / revoke Temporary Head	✗	✗	✓
Appear highlighted as Head	✗	✓ (temp)	✓

8. Constraints & Assumptions

8.1 Constraints

- Google Maps / Distance Matrix API usage must remain within the free-tier quota during development; production billing to be planned.
- WhatsApp Business API requires Meta business verification; the team should apply early as approval can take several days.
- The internship timeline (typically 8–16 weeks) constrains which features reach production-ready state.
- PWA-to-mobile wrapping (Capacitor) is out of scope for v1 but architecture must not preclude it.

8.2 Assumptions

- Target users have a smartphone with a modern browser (Chrome 110+, Safari 16+, Firefox 115+).
- Users grant location permission to the browser/app.
- All group members have a Google account (required for login).
- WhatsApp integration is optional; the app is fully functional without it.
- ETA calculations use current traffic data where available via the Maps API.
- The internship team has access to a Google Cloud project for Maps and OAuth credentials.

9. Acceptance Criteria

The following criteria must be met for v1 to be considered complete:

- AC-01: A user can log in with Google, create a group, and invite at least two other users.
- AC-02: All group members' locations update on the dashboard within 10 seconds of movement.
- AC-03: The Group Head's entry is visually pinned and distinct on all members' dashboards.
- AC-04: The Head can assign and revoke a Temporary Head with a defined expiry.
- AC-05: Distance, locality, and ETA (car + transit) are displayed correctly for at least three test locations.
- AC-06: The 'Call' button triggers a native phone-call intent on mobile devices.
- AC-07: The map drill-down shows profile-picture markers for all members.
- AC-08: A WCAG 2.1 AA automated scan (axe-core) returns zero critical violations.
- AC-09: The Read Aloud feature correctly voices all list items via TTS.
- AC-10: WhatsApp linked users receive a message when the Head's location changes.
- AC-11: A user can link and unlink their WhatsApp account from Account Settings.
- AC-12: All data is deleted within 24 hours of a member being removed from a group.

10. Glossary

Term	Definition
Group Head	The designated administrator and primary reference point of a LocationCircle group.
Temporary Head	A member temporarily elevated to Head status for a defined period.

Term	Definition
LocationCircle	Working title for the application.
PWA	Progressive Web App — a web application installable on mobile home screens.
ETA	Estimated Time of Arrival, calculated via the Google Maps Distance Matrix API.
TTS	Text-to-Speech — the Web Speech API used to vocalise list content for accessibility.
WCAG 2.1 AA	Web Content Accessibility Guidelines version 2.1, Level AA compliance standard.
WhatsApp Business API	Meta's official API for sending and receiving WhatsApp messages programmatically.
MoSCoW	Prioritisation framework: Must have, Should have, Could have, Won't have (this version).